3.1 *F*-DISTRIBUTION AND ITS APPLICATIONS

F-statistic is the ratio of two sums of the squares of deviations of observations from respective means. The sampling distribution of the statistic is *F*-distribution.

Definition: *F*-Distribution

Let X and Y be two independent χ^2 random variates with m and n degrees of freedom respectively. Then $F = \frac{X/m}{Y/n}$ is said to follow *F*-distribution with (m, n) degrees of freedom. This *F*-distribution is named after the famous statistician R.A. Fisher (1890 to 1962).

Definition: *F*-Statistic

Let $(X_1, X_2, \ldots, X_m)$ and $(Y_1, Y_2, \ldots, Y_n)$ be two independent random samples drawn from $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$ populations respectively.

Then,

$$\frac{1}{\sigma_X^2}\sum_{i=1}^{m}\left(X_i-\overline{X}\right)^2 \sim \chi^2_{m-1} \text{ and } \frac{1}{\sigma_Y^2}\sum_{j=1}^{n}\left(Y_j-\overline{Y}\right)^2 \sim \chi^2_{n-1}$$

are independent

(1) Hence, *F*-Statistic is defined as

$$F=\frac{(m-1)S_X^2}{\sigma_X^2}\Big/\frac{(n-1)S_Y^2}{\sigma_Y^2} \sim F_{m-1,n-1}$$

where

$$S_X^2=\frac{1}{m-1}\sum_{i=1}^{m}\left(X_i-\overline{X}\right)^2 \text{ and } S_Y^2=\frac{1}{n-1}\sum_{j=1}^{n}\left(Y_j-\overline{Y}\right)^2$$

(2) *F*-Statistic is also defined as the ratio of two mean square errors.

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If the populations are not normal, *F* – test may not be used.

Assumptions for testing the ratio of two normal population variances

i) The population from which the samples were obtained must be normally distributed.

ii) The two samples must be independent of each other.

Applications of *F*-distribution

The following are some of the important applications where the sampling distribution of the respective statistic under H_0 is *F*–distribution.

(i) Testing the equality of variances of two normal populations. [Using (1)]

(ii) Testing the equality of means of k (>2) normal populations. [Using (2)]

(iii) Carrying out analysis of variance for two-way classified data. [Using (2)]

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